

Total
33
/34

KNOWLEDGE (No Calculators!!!)

1. Which of the following characteristics of the function, $f(x) = \log_{10} x$, changes under the following transformations: $f(x) = -2 \log_{10} 6x$?

- a) The range of the function
b) The domain of the function
c) The x intercept
d) The vertical asymptote

2. The exact value of $\log_3 \sqrt[5]{81}$ is: $(81)^{\frac{1}{5}}$ $(7^4)^{\frac{1}{5}}$

- a) 3
b) 5/4
c) 21/5
d) 4/5

3. Solve for x. Show ALL work for full marks.

a) $x = 2 \log_{\sqrt{2}} (4^2)$

$$x = 2 \log_{2^{\frac{1}{2}}} (2^4)$$

$$x = 2 \log_{2^{\frac{1}{2}}} ((2^{\frac{1}{2}})^3)^4$$

$$x = 2 \log_{2^{\frac{1}{2}}} (2^{\frac{1}{2}})^{12}$$

$$x = 2(12)$$

$$x = 24$$

b) $x = \sqrt{12} \log_{\frac{1}{8}} (2^{\sqrt{3}})$

$$x = \sqrt{12} \log_{2^{-3}} ((2^{-3})^{\frac{1}{3}})^{\sqrt{3}}$$

$$x = \sqrt{12} \log_{2^{-3}} (2^{-3})^{\frac{\sqrt{3}}{3}}$$

$$= \sqrt{12} \left(\frac{-\sqrt{3}}{3} \right)$$

$$= \frac{-\sqrt{36}}{3}$$

$$= -2$$

c) Solve for x, $5 \log_{\sqrt{3}} 81^{3x} = 240$, show all your work.

$$5 \log_{3^{\frac{1}{2}}} (3^4)^{3x} = 240$$

$$5 \log_{3^{\frac{1}{2}}} ((3^{\frac{1}{2}})^2)^{12x} = 240$$

$$5 \log_{3^{\frac{1}{2}}} (3^{\frac{1}{2}})^{24x} = 240$$

$$5(24x) = 240$$

$$120x = 240$$

$$x = 2$$

4. Use mapping rule to graph the following function, $f(x) = -3 \log_{\frac{1}{5}} \left[-\frac{1}{2}x + 5 \right] - 6$. **SHOW ALL WORK.**

Label the function, all intercepts and/or any asymptote(s) on the graph for full marks.

$$f(x) = -3 \log_{\frac{1}{5}} \left[-\frac{1}{2}(x-10) \right] - 6 \quad \frac{48}{5} = 9.6$$

$$(x, y) \rightarrow (-2x+10, -3y-6) \quad -\frac{2}{5} + \frac{56}{5} = \frac{54}{5}$$

$$(1, 0) \rightarrow (8, -6)$$

$$\left(\frac{1}{5}, 1\right) \rightarrow \left(\frac{48}{5}, -9\right)$$

$$VA: x = 10$$

$$f(10) = -3 \log_{\frac{1}{5}}(5) - 6 \quad 0 = -3 \log_{\frac{1}{5}} \left(-\frac{1}{2}(x-10) \right) - 6$$

$$= -3$$

$$-2 = \log_{\frac{1}{5}} \left(-\frac{1}{2}(x-10) \right)$$

$$\left(\frac{1}{5}\right)^{-2} = -\frac{1}{2}(x-10)$$

$$-50 = x-10 \Rightarrow x = -40$$

$$y\text{-int: } (0, -3)$$

COMMUNICATION (No Calculators!!!) $y\text{-int: } (-40, 0)$

5. Michael and Scott are having a conversation about logarithms. Michael says the function $y = \log_{\sqrt{0.5}} x$, does not exist, because the base is not greater than 1. Scott says that does not matter and the function does exist. Who is correct? Explain.

Scott is right in that the base can be less than 1, although greater than 0. (1)

As long as the base is positive, it produces a continuous function; A negative base creates a discontinuous function.

6. Is the following statement True? Justify your answer mathematically. $7 \log_m 100 = \frac{\log 100}{\log_{100} m} \times \sqrt{49}$

$$RS = \frac{\log 100}{\frac{\log m}{\log 100}} \times 7 = 7 \log(100) \log_m(100)$$

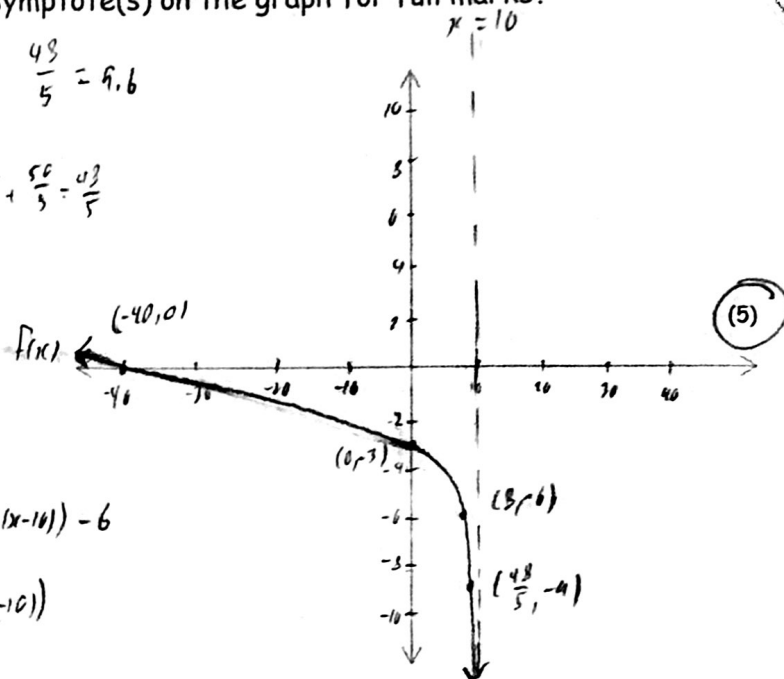
$$= \frac{(\log 100)^2}{\log m} \times 7 \neq LS$$

Not true.

7. Calculate $\sqrt{\frac{6401^2 - 1}{6402}}$

$$= \sqrt{\frac{(6401+1)(6401-1)}{6401+1}}$$

$$= \sqrt{6400} = \pm 80$$



(2)

(1)

Anthony Trang

FORMULAS: $\text{pH} = -\log[H^+]$ $\beta_2 - \beta_1 = 10 \log\left(\frac{I_2}{I_1}\right)$ $M = \log\left(\frac{I_1}{I_0}\right)$

APPLICATION

1. Milk has a pH value of 4.0. How many times more acidic is orange juice, with a pH value of 6.4 than milk?

$$\text{pH} = -\log[H^+]$$

$$6.4 = -\log[H^+]$$

$$4 = -\log[H^+]$$

$$[H^+] = 10^{-6.4}$$

$$[H^+] = 10^{-4}$$

$$\frac{10^{-6.4}}{10^{-4}} = 3.98 \times 10^{-3}$$

$$3.98 \times 10^{-3} \text{ times}$$

(2)

2. The sound level of the crowd at BMO Field is usually 120 dB. The sound level of the last MLS Cup crowd was about 10,000 times more intense than the sound level of the crowd at BMO field. What was the sound level at last year's MLS Cup, in decibels?

$$\beta_{\text{MLS}} - \beta_{\text{BMO}} = 10 \log\left(\frac{I_{\text{MLS}}}{I_{\text{BMO}}}\right)$$

$$\beta_{\text{MLS}} - 120 = 10 \log(10^4)$$

$$\beta_{\text{MLS}} = 160 \text{ dB}$$

$$160 \text{ dB}$$

(2)

3. A Skydiver must time when they open their parachute correctly in order to land on time and in the correct location. A skydivers height is modelled by the function $h(t) = -9 \log(t + 1) + 7$, where $h(t)$ is height in hundred meters and t is time in minutes.

a) How long will it take him to land? Show all work. Round your answer to the nearest minute.

$$0 = -9 \log(t + 1) + 7$$

$$\frac{7}{9} = \log(t + 1)$$

$$t + 1 = 10^{\frac{7}{9}}$$

$$t = 10^{\frac{7}{9}} - 1 \Rightarrow t \approx 5$$

$$5 \text{ minutes}$$

(2)

b) What is the Skydivers initial height? Show all work.

$$h(0) = -9 \log(0 + 1) + 7$$

$$h(0) = -9 \log(1) + 7$$

$$h(0) = 7$$

$$700 \text{ meters}$$

(1)

6. Solve x as an exact solution. Fully simplify the exact solution as much as possible.

$$\frac{1}{2}(4^{-2x+14}) - 6 = 5.5$$

$$\frac{1}{2}(4^{-2(x-7)}) = \frac{23}{2}$$

$$4^{-2(x-7)} = 23$$

$$-2(x-7) = \log_4 23$$

$$x = -\frac{1}{2} \log_4 23 + 7$$

(3)

THINKING

- Graph the following inequality. Label all functions, key features and points of intersection on your graph to receive full marks.

$$-2 \log_2 [2x - 8] + 12 > 3^{x-6} - 3$$

$$f(x) = -2 \log_2 [2(x-4)] + 12$$

$$(x, y) \rightarrow (\frac{1}{2}x + 4, -2y + 12)$$

$$(1, 0) \rightarrow (\frac{9}{2}, 12)$$

$$(2, 1) \rightarrow (5, 10)$$

$$VA: x = 4$$

$$f(6) = -2 \log_2 (8) + 12$$

$$\downarrow$$

$$DNE$$

$$VA: DNE$$

$$0 = -2 \log_2 [2(x-4)] + 12 \quad 2^6 = 2(x-4)$$

$$6 = \log_2 [2(x-4)] \quad x = 2^5 - 4 = 28 \quad x\text{-int}(28, 0)$$

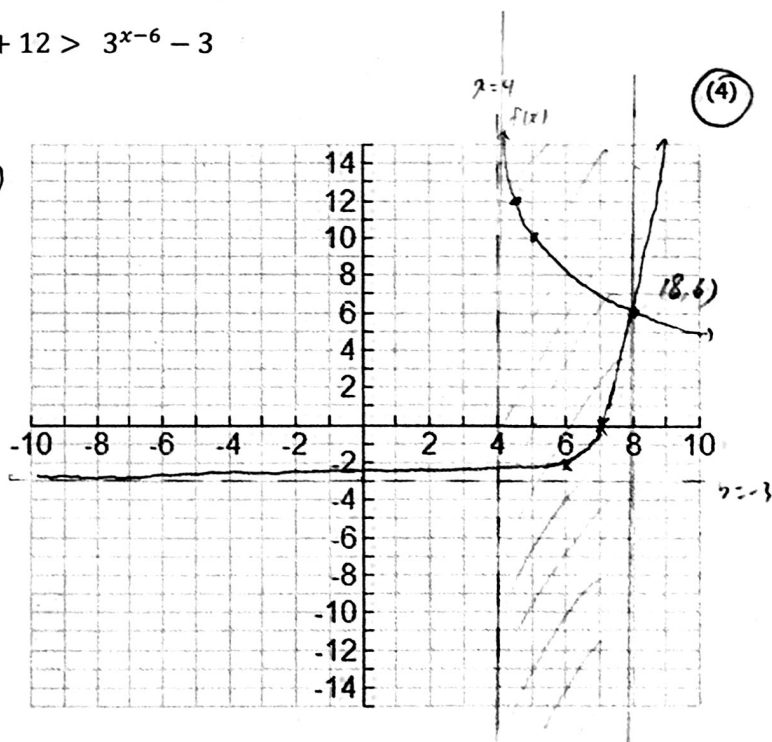
$$g(x) = 3^{x-6} - 3$$

$$(x, y) \rightarrow (x+6, y-3)$$

$$(0, 1) \rightarrow (6, -2)$$

$$(1, 3) \rightarrow (7, 0)$$

$$HA: y = -3$$



(4)

State the solution to the inequality, if possible.

$$\{x \in \mathbb{R} \mid 4 < x < 8\}$$

(11)

$$3^{x-6} - 3 = y$$

$$3^{x-6} = y + 3$$

$$x-6 = \log_3 (y+3)$$

$$x = \log_3 (y+3) + 6$$

$$-2 \log_2 (2 \times 8 - 8) + 12 = 6$$

$$3^{8-6} - 3 = 6$$

$$POI(8, 6)$$